Discrete Mathematics and Its Applications : Kenneth Rosen

- The rules of logic give precise meaning to mathematical statements. These rules are used to distinguish between valid and invalid mathematical arguments.

Propositions

A proposition is a declarative sentence (that is, a sentence that declares a fact) that is either true or false, $\boxed{\text{but not both}}$.
↳ ?

Example 1

All the following declarative sentences are propositions.
1. Washington D.C. is the capital of the United States of America
2. Toronto is the capital of Canada.
3. $1+1=2$ 4. $2+2=3$

Propositions 1 and 3 are true, whereas 2 and 4 are false.

Example 2

Consider the following sentences.
1. What time is it?
2. Read this carefully.

→ Not propositions because they are not declarative sentences.

3. $x+1=2$ 4. $x+y=z$

Sentences 3 and 4 are not propositions because they are neither true nor false. Each of these sentences can be turned into a proposition if we assign values to the variables.

- We use letters to denote $\boxed{\text{propositional variables}}$ (or $\boxed{\text{sentential variables}}$), that is, variables that represent propositions, just as letters are used to denote numerical variables. The conventional letters used for propositional variables are p, q, r, s, ...
- The $\boxed{\text{truth value}}$ of a proposition is true, denoted by T, if it is a true proposition, and the truth value of a proposition is false, denoted by F, if it is a false proposition.
- Propositions that cannot be expressed in terms of simpler propositions are called $\boxed{\text{atomic proposition}}$.